EXP NO: 2
Cables
DATE: 23/7/26

STUDY OF DIFFERENT TYPES OF NETWORK

AIM

To study and understand different types of network cable, To make your own ethernet cross over cable / straight cable

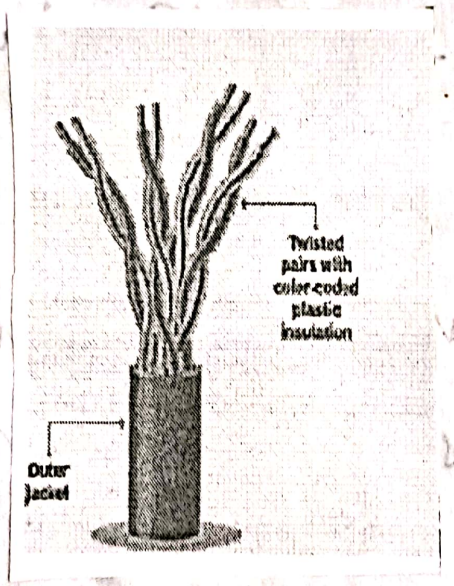
a] Understand different types of network cable

Different types of cable used in networking are:

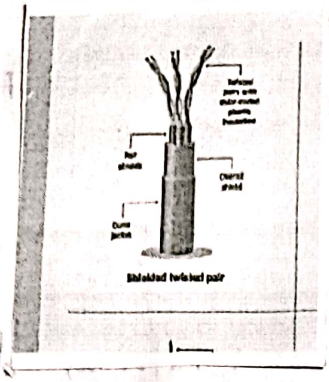
1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. ~~Coaxial~~ Coaxial Cable
4. Fibre optic cable

Cable Type	Category	Maximum Data Transmission	Advantage / Disadvantage	Application use
UTP	Category 3	10 bps	Advantage • cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantage • More prone to electromagnetic interference and noise	10 Base – T Ethernet
	Category 5	upto 100 mbps		Fast Ethernet Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet Gigabit Ethernet

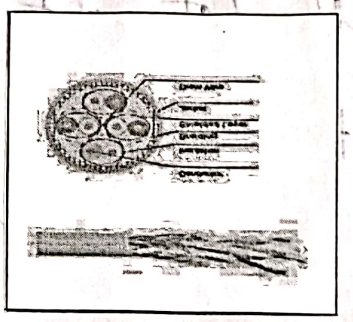
Scanned with OKEN Scanner


Twisted pairs with color-coded plastic insulation
Outer jacket

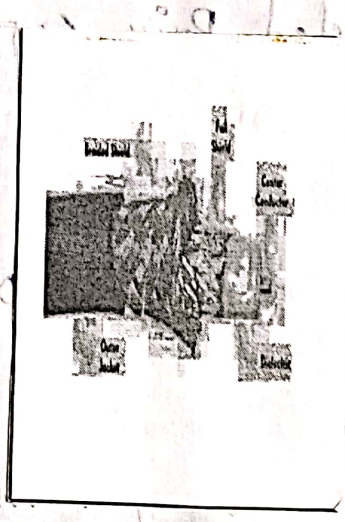
STP

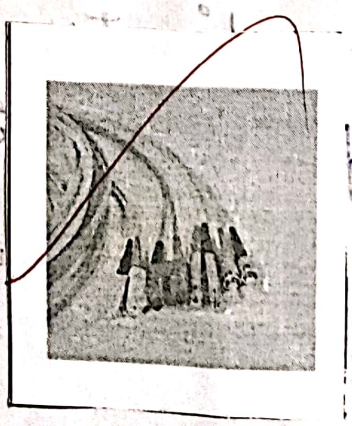


Coaxial cable

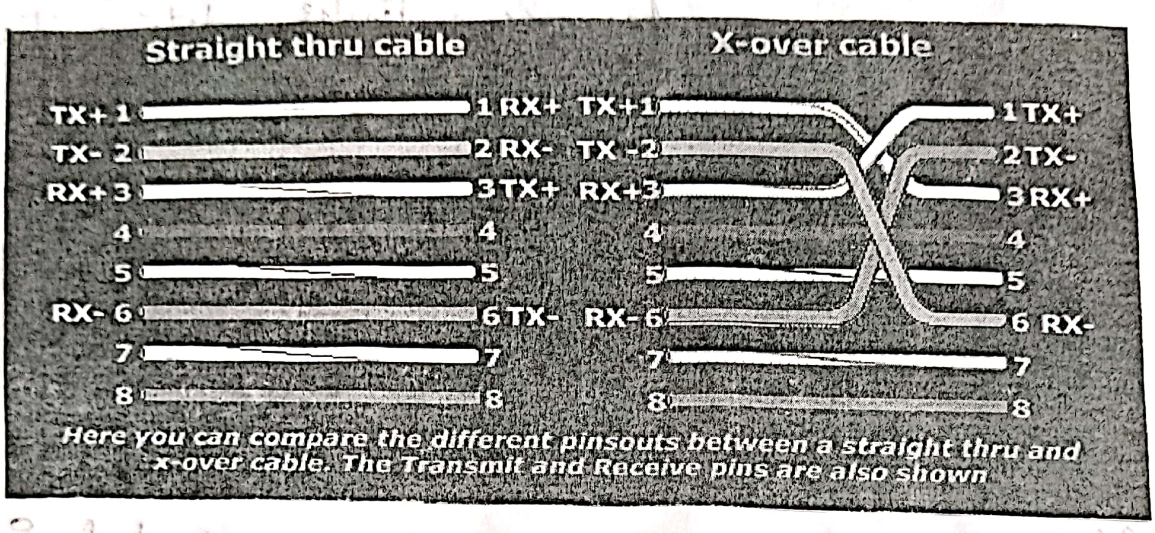


fibre optics cable

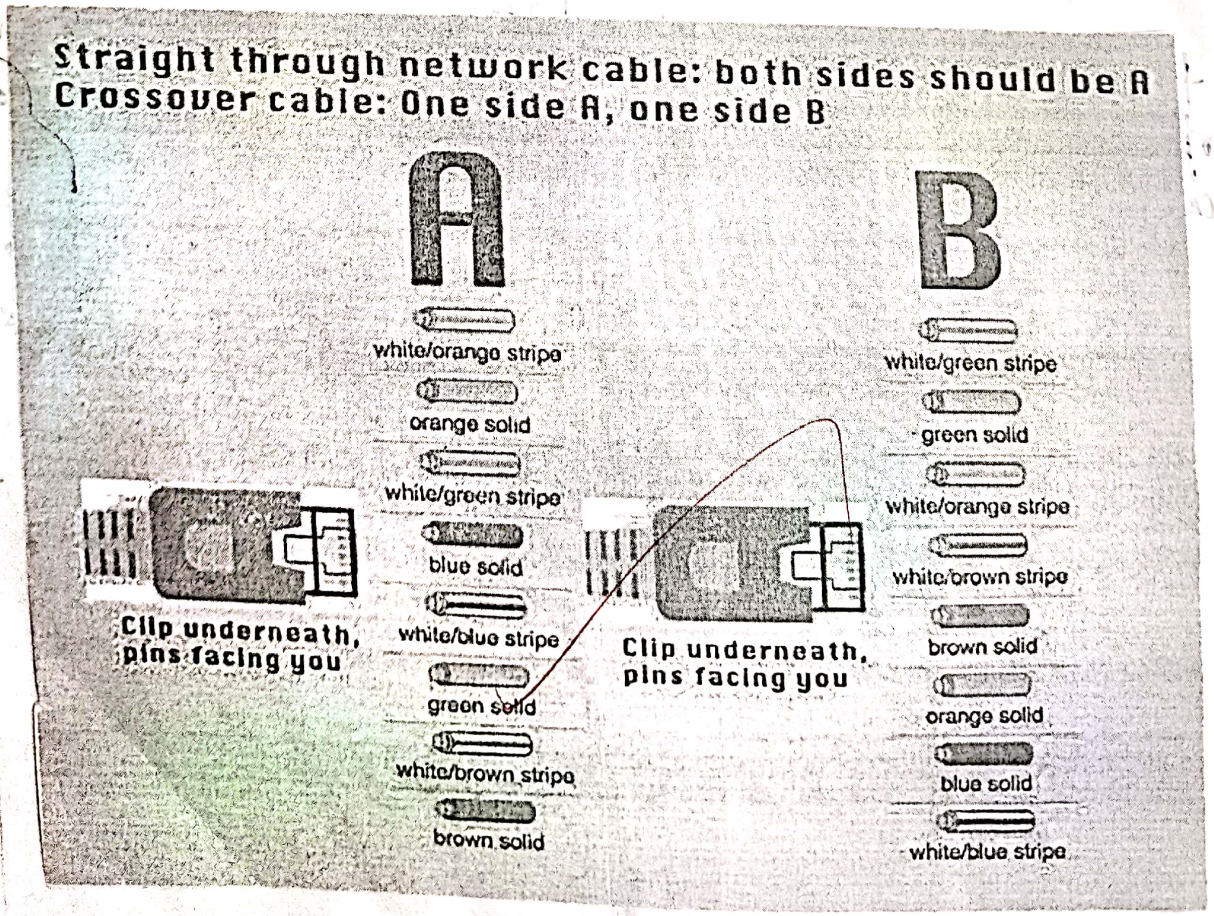




STP	Category6, 6a	10Gbps	Advantages • shielded • faster than UTP • less susceptible to noise and inter ference. Disadvantages • Expensive • Greater installation effort	Gigabit ethernet, 10G ethernet (55m) widely used in data centres Gigabit Ethernet 10G Ethernet (100 m)
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interfere nce • low loss bandwidth • versatile	speed of signal is 500m Television network High speed internet connections
fibre optics cable	single mode multi mode	100 Gbps	Advantages • High speed • High bandwidth • High Security • Long distance Disadvantages • Expensive • Requires skilled installe	• Maximum distance of fibre optics cable is around 100 meters



Difference between crossover cable & straight cable



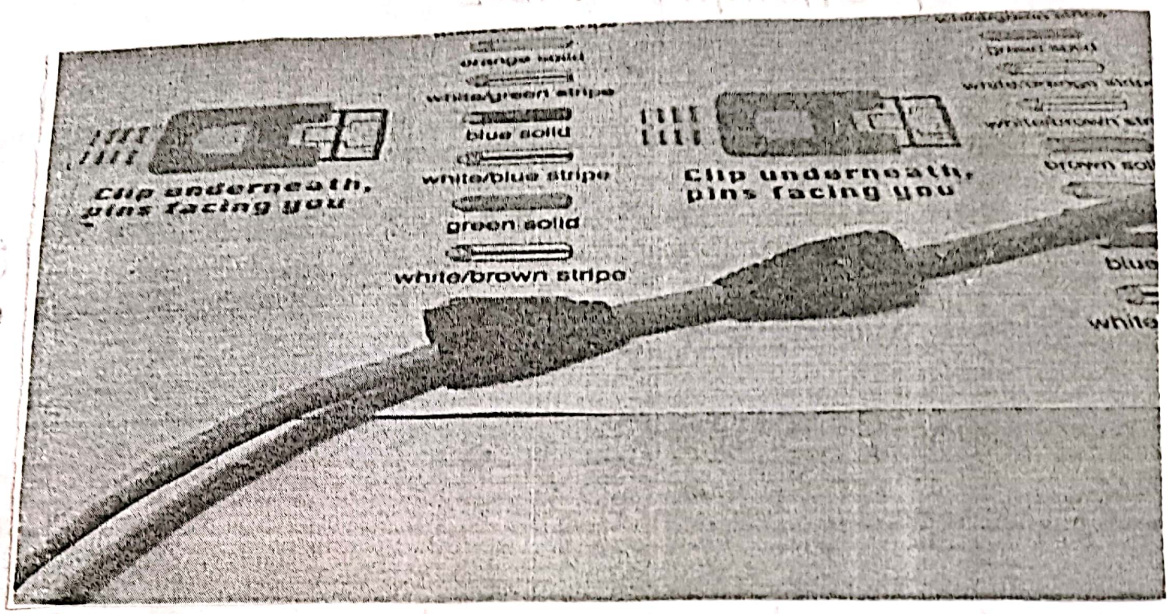
Scanned with OKEN Scanner

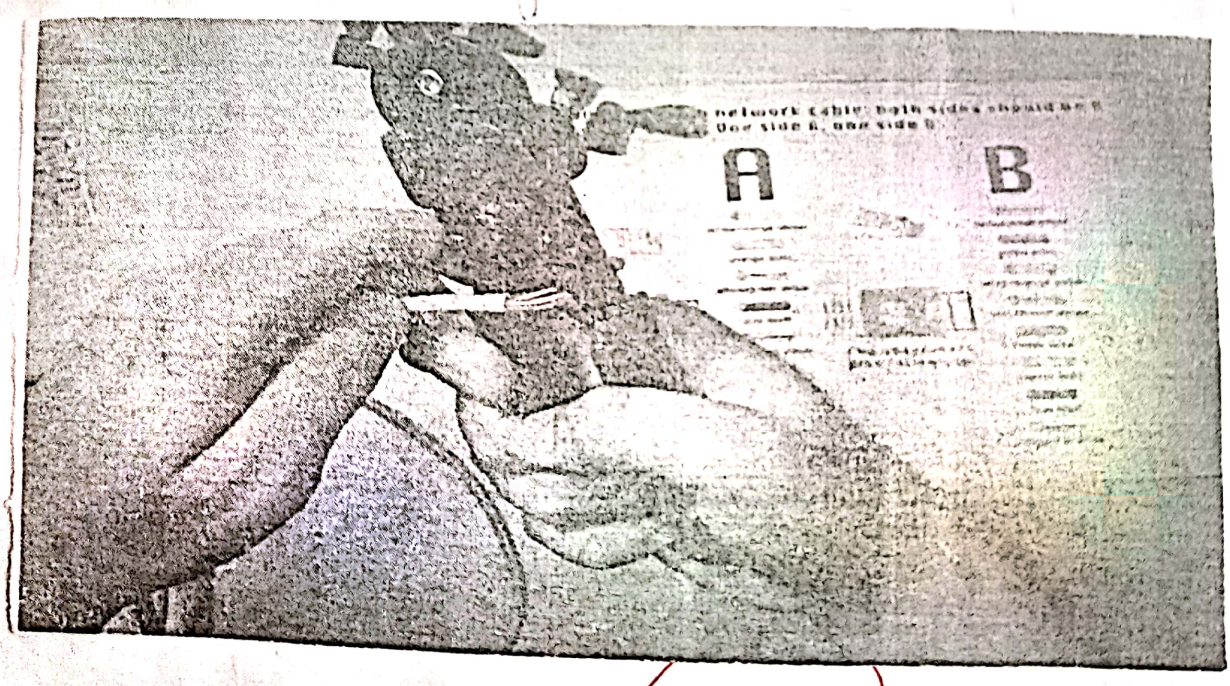
MAKE YOUR OWN ETHERNET CROSS OVER CABLE / STRAIGHT CABLE

Tools and parts needed

- Ethernet cabling. CAT5e is certified for gigabit support but CAT5 cabling works a well just over shorter distances.
- A crimping tool. This is an all in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs
- optional two plug shields.

Scanned with OKEN Scanner


Clip underneath, pins facing you
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you


A
B

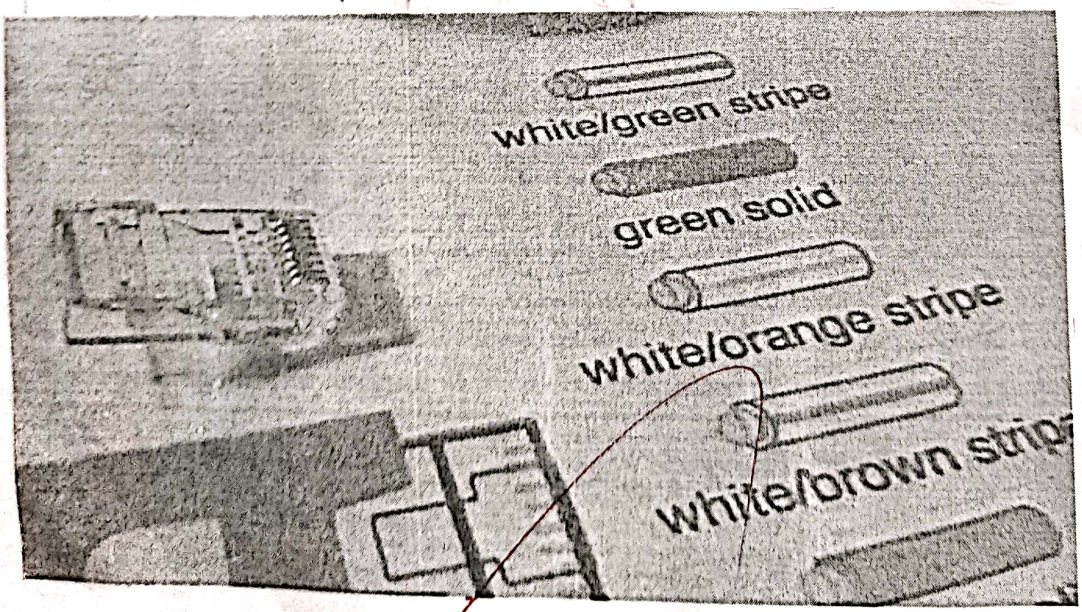
Step 1: To start construction of the device begin ~~the~~ by threading shield onto the cable

~~Step 2: Next~~

Step 2: Next strip approximately 1.5cm of cable shilding from both ends The crimping tool has a round area to complete this task.

Scanned with OKEN Scanner


blue solid

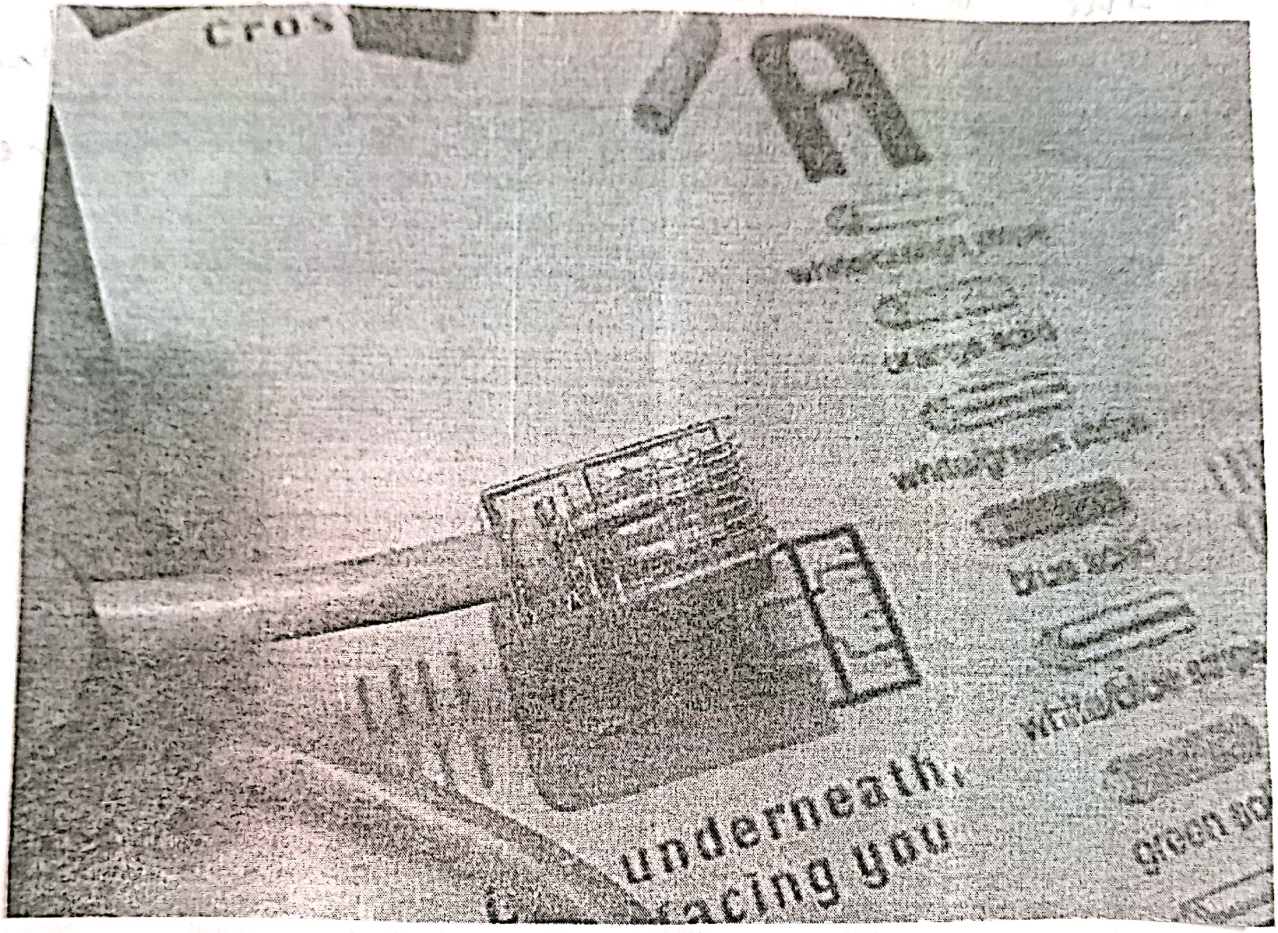

white/green stripe
green solid
white/orange stripe

Step 3: After you will need to untangle the wires there should be four "twisted pairs". Referencing back to the sheet arrange them from top to bottom. One end should be in arrangement A and other in B

~~Step 4:~~

Step 4: Once the order is correct bunch them together in a line and if there are any that stick out farther than others snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug with the clip side facing away from you and have the gold pins facing towards you as shown.

Scanned with OKEN Scanner


underneath
facing you

Step 5: Next push the cable right in. The notch at the end of the plug need to be just over the cable shielding and if isn't that means that you stripped off too much shielding. Simply snip the cables back a little more.

Step 6: After the wires are securely sitting inside the plug insert it into the crimping tool and push down. It should be shaped correctly but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly repeat ~~the~~ for the other end using diagram B (to make a crossover cable) using diagram A (to make a straight through cable)

To test it plug it in and attempt to connect two devices directly.

90

8/1/18

Result:

Thus different type of network cables were studied successfully and straight through and cross over ethernet cables were prepared and tested successfully